

Assignment No. 2

Client.java

```
import java.rmi.*;
import java.util.Scanner;

public class Client {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            String serverUrl = "rmi://localhost/Server";
            ServerInterface serverIntf = (ServerInterface) Naming.lookup(serverUrl);

            System.out.print("Enter a number to find factorial: ");
            int num = sc.nextInt();

            System.out.println("Factorial of " + num + " is: " + serverIntf.factorial(num));

        } catch (Exception e) {
            System.out.println("Exception at Client: " + e);
        }
    }
}
```

Server.java

```
import java.rmi.*;

public class Server {
    public static void main(String[] args) {
        try {
            ServerImplementation serverImpl = new ServerImplementation();
```

```

        Naming.rebind("Server", serverImpl);

        System.out.println("Server is Ready...");
    } catch (Exception e) {
        System.out.println("Exception at Server: " + e);
    }
}
}

```

ServerInterface.java

```

import java.rmi.*;

interface ServerInterface extends Remote {
    public long factorial(int num) throws RemoteException;
}

```

ServerImplementation

```

import java.rmi.*;
import java.rmi.server.*;

public class ServerImplementation extends UnicastRemoteObject implements ServerInterface {

    public ServerImplementation() throws RemoteException { }

    public long factorial(int num) throws RemoteException {
        long fact = 1;
        for(int i = 1; i <= num; i++) {
            fact *= i;
        }
        return fact;
    }
}

```

OUTPUT

Terminal 1

```
C:\Users\anuja>cd C:\Users\anuja\OneDrive\Desktop\LP V\A1
```

```
C:\Users\anuja\OneDrive\Desktop\LP V\A1>rmiregistry
```

Terminal 2

```
C:\Users\anuja\OneDrive\Desktop\LP V\A1>javac *.java
```

```
C:\Users\anuja\OneDrive\Desktop\LP V\A1>java Server
```

```
Server is Ready...
```

Terminal 3

```
C:\Users\anuja>cd C:\Users\anuja\OneDrive\Desktop\LP V\A1
```

```
C:\Users\anuja\OneDrive\Desktop\LP V\A1>java Client
```

```
Enter a number to find factorial: 5
```

```
Factorial of 5 is: 120
```